Robot Design Speech

Write it in your own words. Facts, numbers and reasons.

Name: _____ **Date:** _____

At judging we give a **5-minute robot design explanation** with no game field, then judges ask questions for 5 more minutes. These pages guide you through each part of the speech. They give you sentence starters, not the answers. **You** write what we really did, so it sounds like us.

What to do

1. **Read** the gold box on each page. It tells you what judges listen for.
2. **Write** an answer under each ✎ starter. Use the reference pages (meeting notes, design log, test results) to find facts.
3. **Add numbers.** How many runs? How far off? How many out of 10?
4. **Don't know?** Write **[ASK: name]** and we will track it down.
5. **Practice out loud.** About 600 to 700 words is 5 minutes. Speak slowly.

What makes a speech score high?

- **Say why.** "We changed it because ..." beats "We changed it."
- **Say what is new.** Tell the judges what we changed from the original build.
- **Tell the test story:** we tested, found a problem, changed it, tested again, it got better.
- **Name everyone.** All nine of us speak and all nine of us did real work.
- **Be proud.** Smile. Celebrate what we did.

The robot design process we follow

Identify → Design → Create → Iterate (test and improve) → Communicate what we learned. The seven speech sections follow this order.

What judges listen for

Judges tick one box per row. Clear evidence is **Accomplished (3)**. To score **Exceeds (4)** the judges must write down how we exceeded, so give them something to write.

Area	Accomplished: clear evidence of ...	Exceeds: give them more
IDENTIFY	Mission strategy. Use of building and coding resources.	Why we chose missions in that order. Which resources, and what we took from each. Who we asked for help.
DESIGN	All members contributed ideas. Building and coding skills in all members.	Name **every** teammate and one thing each did. Who learned a new skill?
CREATE	Clear explanation of attachments and their purpose. Clear explanation of code and sensors.	What we **changed from the original** and why. Code explained **step by step**. What each sensor does for us.
ITERATE	Repeated testing. Improvements based on testing.	**Numbers** (how many runs, how far off). A story: test, problem, change, test again.
COMMUNICATE	Detailed process and lessons learned. Pride and enthusiasm.	Advice for another team. A mistake we learned from. Smiles and teamwork.

Who says what

Everyone speaks: Sylvie, Brynn, Jack, Fedor, Ira, Jordan, Ella, Alexis, Amelia. Times add up to about 4 minutes 40 seconds. You can split a section between two people.

Section	Time	Speaker(s)
1. Opening and mission strategy	0:30	
2. Choosing and building our robot	0:40	
3. Our attachments	1:00	
4. Our code and sensors	0:50	
5. Testing and improving	0:50	
6. What we learned	0:25	
7. Teamwork and closing	0:25	

1 Opening and mission strategy

IDENTIFY

WHAT JUDGES WANT

How we decided which missions to try, and which resources and people we learned from. Name at least three resources and say what each gave us.

✎ **Hello, we are team _____ and today we will explain how we designed our robot. Our strategy is to attempt these missions:**

✎ **We chose those missions because ...** (points? location? one attachment for several missions? Mission 11 is worth 30 points)

✎ **We decided to do them in this order because ...**

✎ **To learn building and coding we used these resources: (bot-fluencers, conference sessions, SPIKE app, build instructions)**

✎ **From _____ we learned _____, and we used it for _____.**

✎ **When we got stuck we asked _____ for help with _____.**

2 Choosing and building our robot

IDENTIFY + DESIGN

WHAT JUDGES WANT

A reason for our base robot, not just its name. Facts: we scored 4 designs on 6 criteria about match performance, left cost out on purpose, and HummerOne scored 23 out of 30. See log entries 1, 2, 3, 5 and 9.

✎ **We looked at ___ designs: _____. We scored them on ___ things: _____.**

✎ **We left out _____ on purpose because ...**

✎ **The highest score was _____ with ___ out of ___. We chose it because ...**

✎ **After building it we noticed this problem:** (front wheels? something sticking out?)

✎ **So we changed _____ to _____ because ... (then again, if it changed twice)** (what was wrong with each version?)

3 Our attachments (part 1)

CREATE

WHAT JUDGES WANT

For each attachment: what it does, what we changed from the original, and why. Do not only say what it is. Say what is new.

✎ **We built ___ attachments. They all attach to the robot's _____ using _____.**

✎ **Our attachments swap on and off in about ___ seconds. We tried _____ pins to make it faster, but ... so we went back to _____ because ...** (log entry 7)

✎ **We rated our attachments from 1 to 5 and chose _____ because ...**

Attachment: _____ **Mission(s):** _____

✎ **It started as a design from:**

✎ **What we changed from the original, and why:**

✎ **A problem we found, and how we fixed it:**

✎ **It is good at ... and not so good at ...**

Our attachments (part 2)

One box per attachment. Choose from: lifting arm with seed bucket, 2-axis arm with hook, forklift, Mission 1 attachment, Mission 11 basket.

Attachment: _____ **Mission(s):** _____

✎ **It started as a design from:**

✎ **What we changed from the original, and why:**

✎ **A problem we found, and how we fixed it:**

✎ **It is good at … and not so good at …**

Attachment: _____ **Mission(s):** _____

✎ **It started as a design from:**

✎ **What we changed from the original, and why:**

✎ **A problem we found, and how we fixed it:**

✎ **It is good at … and not so good at …**

Our attachments (part 3)

Keep going until every attachment we will talk about has a box.

Attachment: _____ **Mission(s):** _____

It started as a design from:

What we changed from the original, and why:

A problem we found, and how we fixed it:

It is good at ... and not so good at ...

Attachment: _____ **Mission(s):** _____

It started as a design from:

What we changed from the original, and why:

A problem we found, and how we fixed it:

It is good at ... and not so good at ...

4 Our code and sensors (part 1)

CREATE

WHAT JUDGES WANT
Explain it step by step, as if the judge has never coded. Say what each sensor does for us.

✎ **We use ___ sensors on our robot: _____. The sensor in the _____ does _____, and we use it to _____.**

✎ **We added a second color sensor because ...**

✎ **Squaring up on a line, step by step:** (with one sensor it did not work. With two, what happens when one sees black?)

First, ...

Then, ...

Finally, ...

4 Our code and sensors (part 2)

CREATE

WHAT JUDGES WANT
Gyro Move, Kp and Ki, gyro turns, the speed ramp and the centimetre conversion. See log entries 8 and 10.

✎ **To drive straight our code uses a block called _____. Every 0.1 seconds the robot checks _____ and if it is off it ...**

✎ **Kp decides _____. Ki decides _____.**

✎ **For turning we use _____ instead of motor degrees because ...** (yaw 90 versus motor 90)

✎ **The robot starts slowly, speeds up and slows at the end because ...**

✎ **We changed the distance block from _____ to _____ because ... We measured that ___ motor degrees is ___ cm by ...**

5 Testing and improving (part 1)

ITERATE

WHAT JUDGES WANT
This is where teams exceed: **repeated** testing and improvements **because of** the testing. Use numbers. Pick the story you have the best evidence for.

Ideas: ☐ Kp and Ki tuning ☐ 1 m drive, 10 times ☐ seed basket (3, 5, 7 out of 10) ☐ front wheel changes ☐ quick versus axle pins ☐ forklift tipping ☐ squaring with two sensors

✎ **Story 1. We wanted to know ...**

✎ **So we tested by _____, ___ times. The results were: (give numbers)**

✎ **That told us ... So we changed ...**

✎ **After the change we tested again and ...**

5 Testing and improving (part 2)

ITERATE

WHAT JUDGES WANT

A second test story with numbers. Then be honest about what is not finished and say your plan.

✎ **Story 2. We wanted to know ...**

✎ **So we tested by _____, ___ times. The results were: (give numbers)**

✎ **That told us ... So we changed ...**

✎ **After the change we tested again and ...**

✎ **Not finished yet: we still need to _____, and our plan is to _____.** (saying this sounds confident, not weak)

6 What we learned

COMMUNICATE

WHAT JUDGES WANT
A detailed explanation of the process and the lessons we learned. Mistakes that taught us something are good here.

The biggest thing we learned about building a robot is ...

The biggest thing we learned about coding is ...

Something we thought would work but did not was _____. We learned ...

If we were giving advice to a new team we would tell them ...

If we had more time we would ...

7 Teamwork and closing

DESIGN + COMMUNICATE

WHAT JUDGES WANT
All team members contributed ideas, with building **and** coding skills in all members. Plus pride and enthusiasm, and a Core Value.

Name	One thing I built or designed	One thing I coded, tested or learned
Sylvie		
Brynn		
Jack		
Fedor		
Ira		
Jordan		
Ella		
Alexis		
Amelia		

✎ **Core Value moment (teamwork, discovery, inclusion, innovation, impact or fun): when _____ ...**

✎ **The part of our robot we are proudest of is _____ because ...**

✎ **Thank you for listening. We would love to answer your questions about ...**

Before we practice

	Checklist
☐	Every blank has a fact, a number or an [ASK: name]
☐	Every choice has a "because"
☐	All 9 teammates speak
☐	At least 2 test stories with numbers
☐	For each attachment: what we changed from the original
☐	Code explained step by step
☐	A lesson from a mistake, and a Core Value
☐	Read out loud, under 5 minutes, without the field and without reading from the page

Questions judges may ask

Jot a short answer or the facts you will use.

1. Why did you choose this base robot?

2. What is the most important change you made, and why?

3. How do you know your robot is accurate?

4. Walk us through how your code works.

5. What did you do when something did not work?

6. Who built each attachment? Who wrote the code?

7. How did you choose which attachment for each mission?

8. What would you change with more time?